Low-pass vs High-pass Filtering of Sentinel-2 Imagery

Which technique suits which remote sensing task?

EES403 Course Project

Group 1

Pranay Chakraborty (MS22041), Sangeeta Kumari (PH25084), Aparna Manoj (MS24070), Robin (MS23138), Sangam (MS23222)

Instructors: Dr. Chandrakant Ojha and Dr. Meenakshi Devi

Department of Earth and Environmental Sciences

Indian Institute of Science Education and Research, Mohali



Spatial frequency: how fast brightness changes across the ground

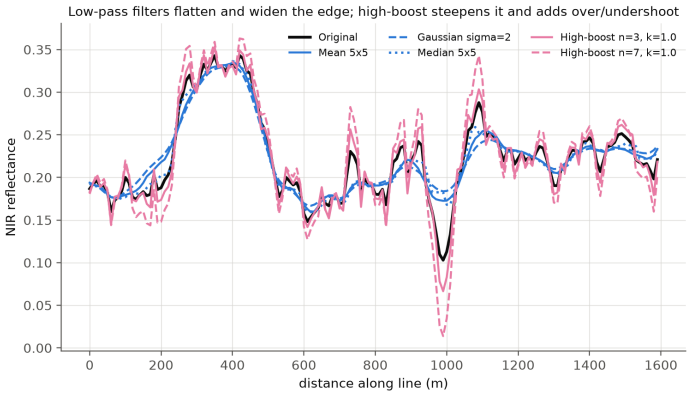
Low frequency: slow change
fields, water, broad slopes

High frequency: abrupt change
roads, roofs, ridge crests, noise

Low-pass (LPF): keep slow change
= smoothing

High-pass (HPF): keep abrupt change
HPF = original − LPF

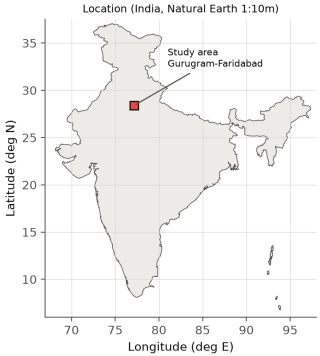
High-boost: original + k·HPF
= sharpening

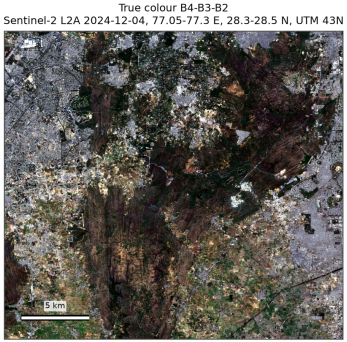


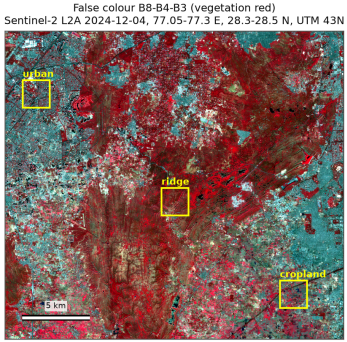
NIR reflectance along a 1.6 km line on the ridge.

Our question: which of the two is more useful, and *for which job*?

Study area: ridges, city and farmland in one scene



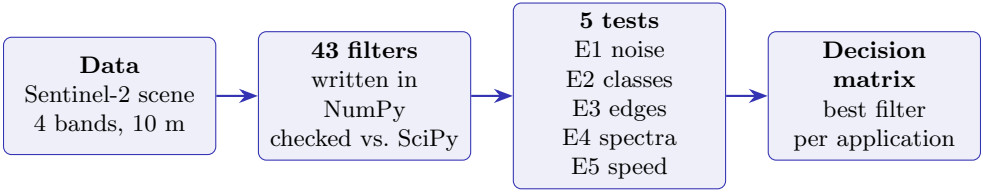




Area: Aravalli ridge, Gurugram–Faridabad, 25 × 23 km
Bands: blue, green, red, NIR at 10 m

Scene: Sentinel-2 L2A, 2024-12-04, best of 11
Reference: ESA WorldCover 2021 land cover

What we did



Spatial domain: slide a kernel

$g(x, y) = \sum_{s,t} w(s, t)\, f(x - s,\, y - t)$

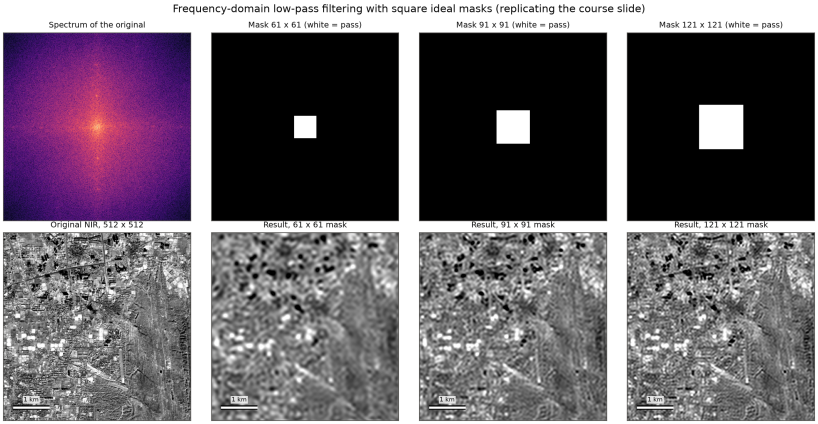
mean, Gaussian, median; HPF, Laplacian, Sobel, Prewitt; high-boost

Frequency domain: mask the spectrum

$g = \mathcal{F}^{-1}\{H(u, v) \cdot \mathcal{F}\{f\}\}$

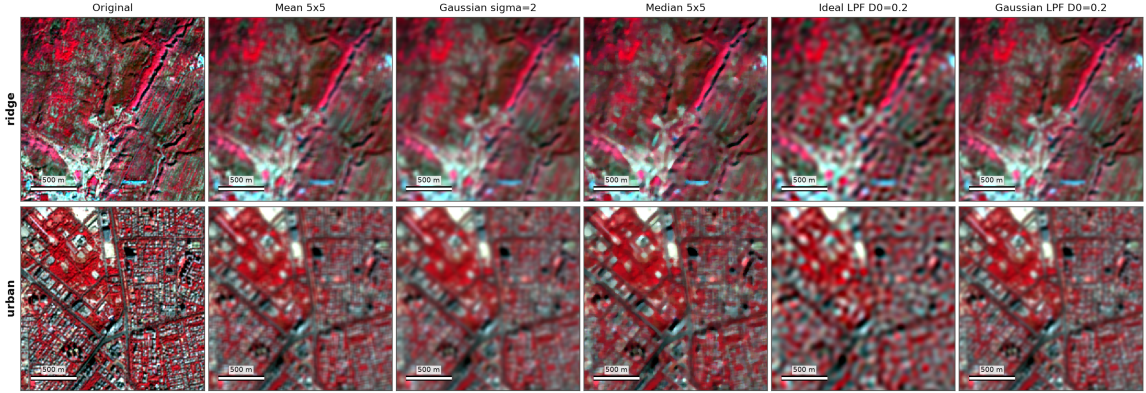
ideal, Butterworth, Gaussian masks; HPF mask $= 1 - H$

Frequency domain: the lecture example, reproduced on our scene



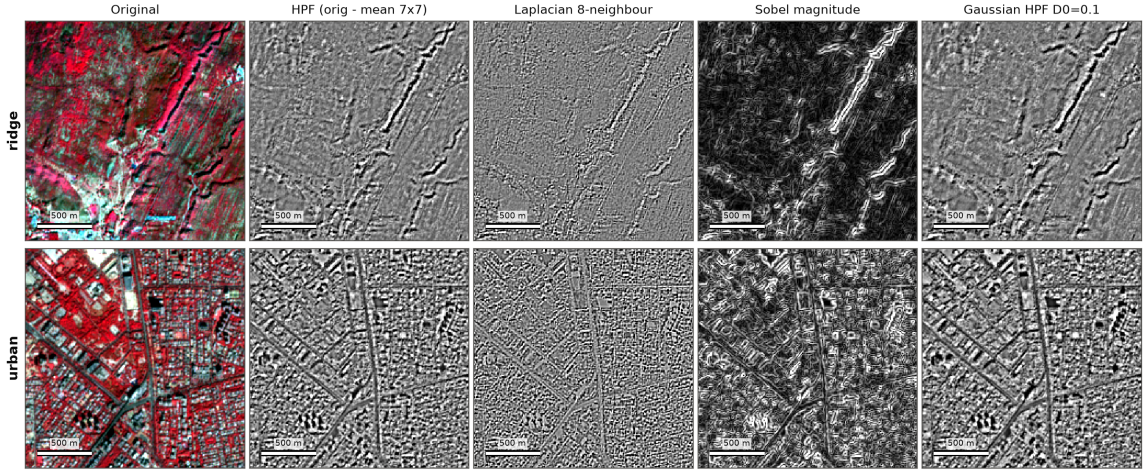
Smaller mask ⇒ fewer frequencies kept ⇒ blurrier image, with ripples

Low-pass filters smooth texture and blur boundaries



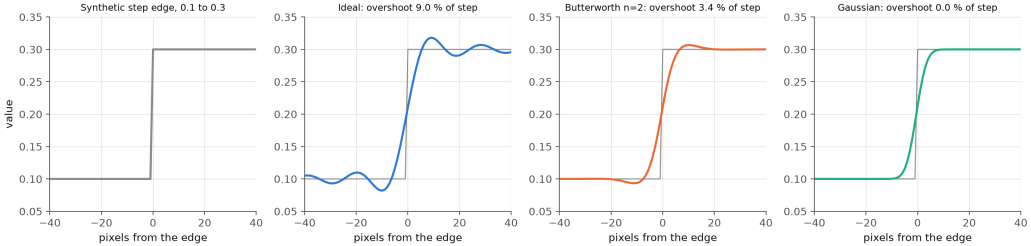
Broad ridge shape survives; street grid and field edges are lost. Median keeps edges best.

High-pass filters keep only the edges



Flat ground turns grey. Sobel and the 7×7 HPF trace ridge crests; the Laplacian picks up speckle.

A sharp cut in frequency causes ringing



Mask ($D_0 = 0.1$)	Ideal	Butterworth	Gaussian
Overshoot at a step edge	**9.0%**	3.4%	0.0%

Never use the ideal mask. Gaussian or Butterworth masks are safe.

E1 Noise: the right filter depends on the type of noise

Random (Gaussian) noise

Noisy input	33.98 dB
Gaussian $\sigma = 1$	**35.56 dB**
Mean 3×3	35.52 dB

13 of 18 filters made it *worse*

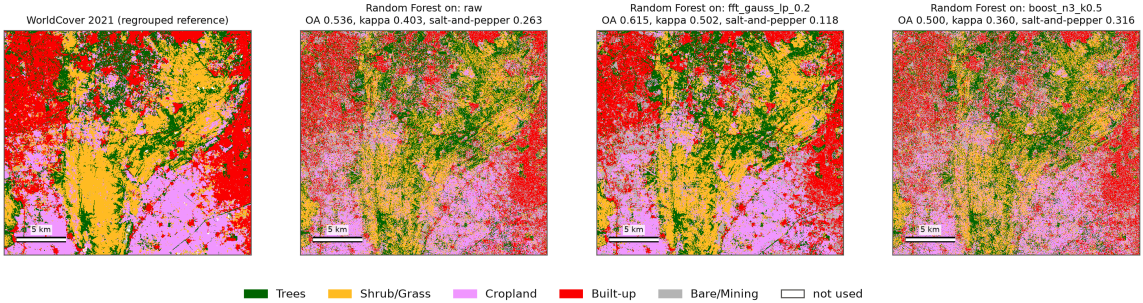
Salt-and-pepper noise

Noisy input	21.01 dB
Median 3×3	**36.49 dB**
Gaussian LPF $D_0 = 0.2$	31.06 dB

median beats the best linear filter by 5.42 dB

PSNR against the clean image, higher is better. Light filters only: large kernels remove detail along with noise.

E2 Classification: smoothing gives more accurate, cleaner maps



Cohen's κ on held-out blocks: raw 0.403 $\rightarrow$ **best LPF 0.502**; best high-boost 0.360

Smoothing removes within-class variation (roof types, shadows) before the classifier sees it.

E3, E4: high-pass finds edges, but the output is no longer reflectance

Edges and lineaments (NIR)

	Original	Best
Boundary F1	0.675	**0.742** high-boost
Ridge coherence	0.510	**0.536** Prewitt

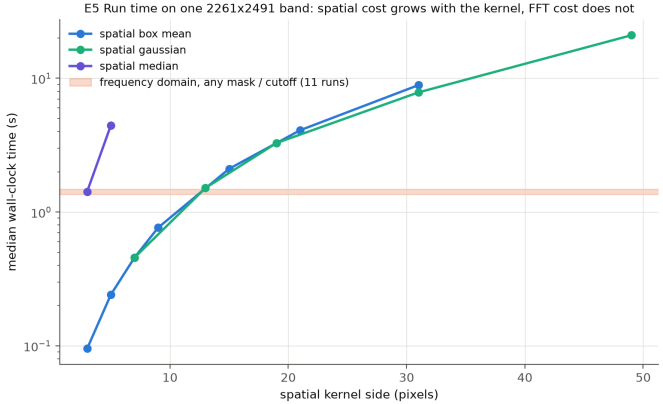
Smoothing lowers boundary F1 (0.247 for mean 5×5)

Cost: change to the spectrum

	Spectral angle
High-boost $n = 3$, $k = 0.5$	1.20°
Median 3×3	2.22°
Gaussian $\sigma = 1$	2.30°
Pure HPF layers	$\approx 89^\circ$

Use HPF products as *extra layers* beside the bands, never in place of them.

E5 Speed: FFT cost is flat, convolution cost grows with the kernel



One full band, 2261×2491 pixels:

- Spatial 3×3: 0.10 s
- Spatial 49×49: 20.96 s
- Any FFT mask: 1.36–1.48 s

Break-even near a 13-pixel kernel

Small kernels: convolve. Large kernels or many cutoffs: use the FFT.

The answer: one best filter per application

Application	Best filter	Evidence (best vs. no filter)
Noise removal, random noise	Gaussian $\sigma = 1$	PSNR (dB): **35.56** vs 33.98
Noise removal, impulse noise	Median 3×3	PSNR (dB): **36.49** vs 21.01
Land-cover classification	Gaussian LPF $D_0 = 0.2$	Cohen's κ: **0.502** vs 0.403
Lineament mapping (ridges)	Prewitt	ridge coherence: **0.536** vs 0.510
Urban boundary delineation	High-boost $n = 3$, $k = 0.5$	built-up edge F1: **0.545** vs 0.543
Visual interpretation	High-boost $n = 3$, $k = 1.0$	NIR mean gradient: **0.0417** vs 0.0245

Low-pass when pixel values feed a calculation; **High-pass** when edges and structure are the point.

Take-home message

Quantitative work

classification, indices, change detection

$\Rightarrow$ light **Gaussian** ($\sigma \approx 1.5$–2 px)
$\Rightarrow$ **median** 3×3 for impulse noise

Interpretation and mapping

visual analysis, lineaments, boundaries

$\Rightarrow$ **high-boost** $k = 0.5$–1 for display
$\Rightarrow$ **Sobel / Prewitt** for lineaments

Avoid

the ideal frequency mask (ringing), and heavy smoothing (detail goes before noise)

Live demo: `python -m rsfilter demo` (offline, 512×512 sample, about one minute)

Thank you. Questions?

References

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